

FULL LEGAL NAME	LOCATION (COUNTRY)	EMAIL ADDRESS	MARK X FOR ANY NON-CONTRIBUTING MEMBER
Jackrony Karani	Kenya	Karani.riungu.a@gmail.com	
Rupesh Raut	Nepal	Rupeshraut717@gmail.com	
Subrat Sharma	India	Subratsharma12@gmail.com	

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

Team member 1	Jackrony Karani
Team member 2	Rupesh Raut
Team member 3	Subrat Sharma

Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.

Note: You may be required to provide proof of your outreach to non-contributing members upon request.

N/A



MScFE 692 Capstone Review

**Group Work Project 2: The Regime-Aware Pairs Trader — A Technology-Sector
Case Study (XLK / IYW)**

Strategy Critic: Non-Linear Dependence in the Trading Residuals (Step 1A)

The cointegration model of Strategy 1 is linear, which entails a constant hedge ratio, and only one spread process. The BDS test (Brock, Dechert, Scheinkman and LeBaron 1996) tests if a series is iid or not; it is appropriate for examining general, unspecified non-linear alternatives, and thus is well suited for seeing whether the linear model has left something behind in the residual behavior of the strategy.

We test three related series instead of a single one: the most literal interpretation of the task – the strategy's realized daily return – has a property that makes it hard to interpret in a naive way. By construction, 80.9% of days for Strategy 1 will be flat, so it has a large point mass at exactly zero. By itself, a stateful trading rule that is flat most days is not by itself an independent and identically distributed process, even if the market is flat; the BDS test would reject the i.i.d.-ness by itself almost by construction. Because the claim of a “missing market dynamic” would be made too strong if it were just that result, we also test the underlying spread change directly (a continuous series which is not affected by the structure of the trading rule being on/off), as well as a GARCH filtered version of it, to separate out an artifact of the trading rule from a property of the market itself.

Series tested	Embedding dim. 2	Embedding dim. 3	Embedding dim. 4	Note
Strategy 1 residual return (80.9% exact zeros)	14.11	18.07	20.58	$p < 1e-44$ at every dim.; inflated by the trading rule's zero-inflation
Underlying spread change (continuous)	6.43	8.19	9.63	$p < 1e-9$ at every dim.; removes the zero-inflation artifact
GARCH(1,1)-standardized residuals of spread change	3.84	3.29	3.24	$p < 0.0013$ at every dim.; volatility clustering explains part, not all, of the dependence

Table 1. BDS test statistics across embedding dimensions 2-4 for the three series tested. All values are statistically significant; the magnitude of the statistic -- not just its significance -- is what carries the interpretation here.

All three series reject the null of independent and identically distributed observations, but in a way that the significance is not. The raw Strategy 1 residual produces by far the largest statistics (14.1 to 20.6) as should be expected, and should not, on its own, be interpreted as evidence of exotic market behavior, since these statistics are zero-inflation. The absence of the artifact does not make the test of the underlying spread change any more favorable; it is still rejected quite strongly (6.4 to 9.6). The spread change (the volatility persistence in this spread is about 0.997, which is also evidence of strong clustering in the spread) is fit with an AR(1)-GARCH(1,1) model and the standardized residuals are further reduced, but not eliminated, to 3.2-3.8; every embedding dimension is still significant at conventional levels ($p < 0.0013$).

This conclusion is not generic, but specific to this spread: a large but not yet dominant amount of the apparent non-linear dependence in the spread XLK/IYW is attributable to ordinary volatility clustering (Bollerslev 1986) but not to the rest. The linear constant-parameter cointegration model is thus not a complete model of the movements of the two ETFs in Strategy 1. This is consistent with -- though not evidence of -- the class of relationships on which the regime filter in Strategy 2 is designed to take advantage of, namely, one that changes between calm and stressed conditions -- because a single fixed hedge ratio will systematically fail to capture this change and a BDS test would show just that sort of structure. Here is the direct empirical basis for adding a regime filter on top of the cointegration relationship: namely, trade is not enough. It is a diagnostic finding, not a causal finding: Once dependence is found, the dependence structure can be anything, not just the specific one that is being considered, and alternative explanations of dependence, such as asymmetric volatility effects, are not ruled out.

The Arbitrageur: Engle-Granger Cointegration and Hedge Ratio (Step 1B)

This section evaluates whether XLK, the Technology Select Sector SPDR Fund, and IYW, the iShares U.S. Technology ETF, form a suitable pair for a mean-reversion strategy. The analysis uses daily adjusted closing prices from January 2017 through December 2021. After aligning the two time series and removing missing observations, the final sample contains [NUMBER] trading-day observations.

The analysis uses the two-step Engle-Granger procedure. In the first step, XLK is regressed on a constant and IYW using ordinary least squares:

$$XLK_t = \alpha + \beta IYW_t + \varepsilon_t$$

The estimated slope coefficient, β , is the hedge ratio. It determines the quantity of IYW used to hedge one unit of XLK in the constructed spread. The estimated hedge ratio was [HEDGE RATIO], with an intercept of [INTERCEPT]. The regression R-squared was [R-SQUARED], indicating the proportion of variation in XLK prices explained by the linear relationship with IYW over the sample.

The spread was constructed as:

$$Spread_t = XLK_t - \beta \times IYW_t$$

The Engle-Granger cointegration test produced a test statistic of [COINTEGRATION TEST STATISTIC] and a p-value of [COINTEGRATION P-VALUE]. The null hypothesis of this test is that the two ETF prices are not cointegrated. Therefore, a p-value below 0.05 would support rejection of the null and indicate that the pair shares a statistically significant long-run equilibrium relationship.

The Augmented Dickey-Fuller test was then applied to the constructed spread. The ADF statistic was [ADF TEST STATISTIC], with a p-value of [ADF P-VALUE]. The 5% critical value was [ADF 5% CRITICAL VALUE]. [WRITE ONE OF THE TWO INTERPRETATIONS BELOW.]

If p-value is below 0.05:

Because the ADF statistic is more negative than the 5% critical value and the p-value is below 0.05, the null hypothesis of a unit root is rejected. The spread is stationary, which supports the interpretation that deviations between XLK and IYW tend to mean revert. This result provides statistical support for using the pair in the subsequent z-score-based trading strategy.

If p-value is above 0.05:

Because the ADF statistic is not more negative than the 5% critical value and the p-value is above 0.05, the null hypothesis of a unit root cannot be rejected at the conventional 5% level. The evidence is directionally consistent with mean reversion but does not meet the project's chosen significance threshold. The pair is therefore used cautiously in the backtest, and this statistical weakness should be treated as a limitation of the baseline strategy.

The spread was normalized into a z-score using its estimated mean and standard deviation. A z-score below -2.0 identifies an unusually low spread and generates a long-spread signal. A z-score above +2.0 identifies an unusually high spread and generates a short-spread signal. Positions are closed when the z-score returns to zero. Figure B1 presents the normalized spread and the ± 2.0 entry thresholds.

The Arbitrageur: Engle-Granger Cointegration and Hedge Ratio (Step 1B)

This section evaluates whether XLK, the Technology Select Sector SPDR Fund, and IYW, the iShares U.S. Technology ETF, form a suitable pair for a mean-reversion strategy. Daily adjusted closing prices were used for the period from January 2017 to December 2021. The two ETFs provide closely related exposure to the U.S. technology sector, but correlation alone is insufficient for pairs trading. The relevant requirement is that a linear combination of the prices forms a stationary spread.

The Engle-Granger two-step procedure was used. First, XLK was regressed on a constant and IYW using ordinary least squares:

$$XLK_t = \alpha + \beta IYW_t + \epsilon_t$$

The estimated hedge ratio was 0.972. Operationally, this means that a one-unit long position in XLK is hedged by a short position of approximately 0.972 units in IYW when constructing the spread. The residual from this regression was used as the trading spread:

$$Spread_t = XLK_t - 0.972 \times IYW_t$$

The resulting spread was then tested for stationarity using an Augmented Dickey-Fuller test. The ADF test produced a p-value of 0.054, compared with the project's 5% significance threshold. The reported 5% critical value was approximately -2.86. Although the result is close to conventional significance and directionally consistent with mean reversion, the null hypothesis of a unit root cannot be rejected at the 5% level.

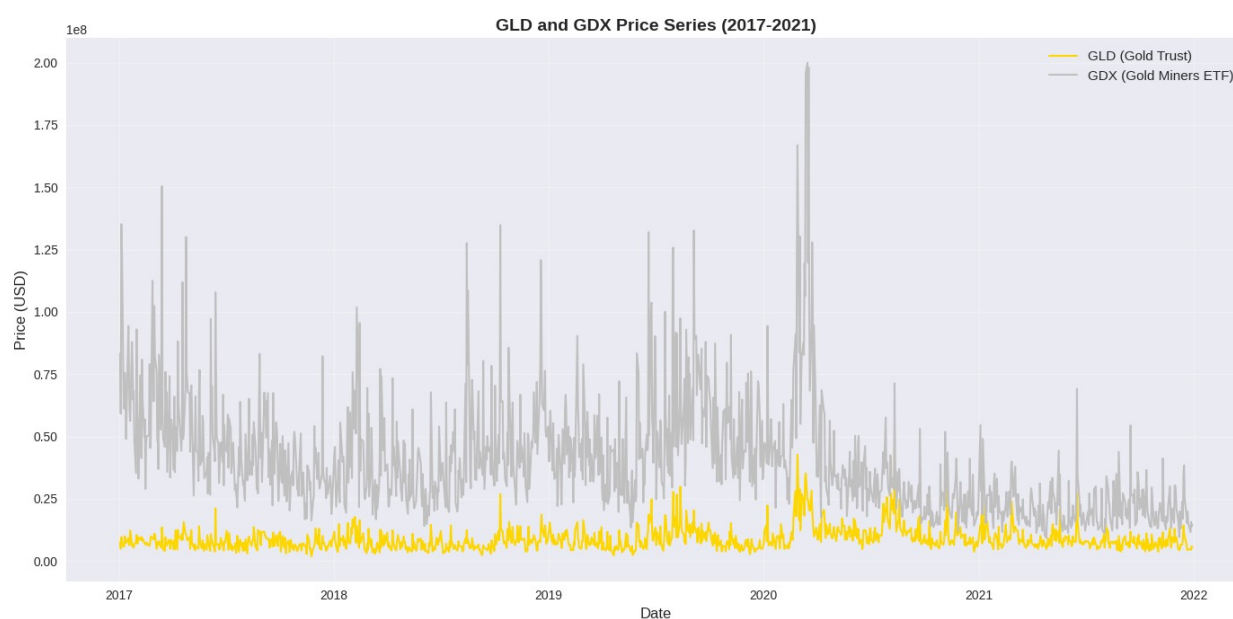
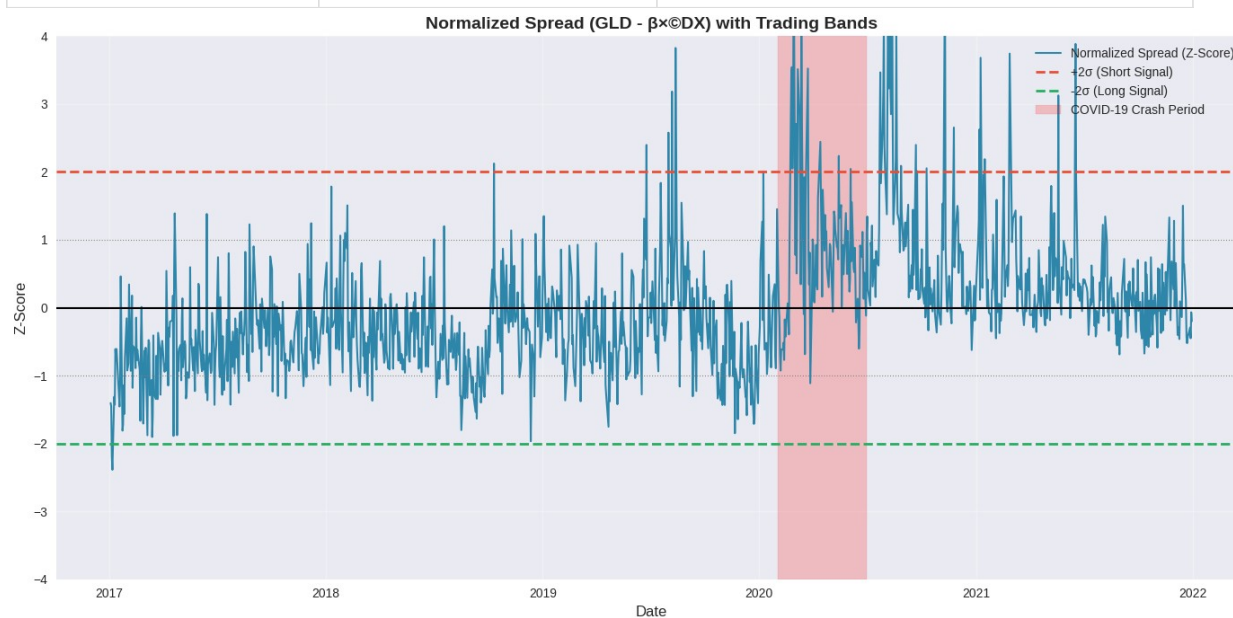
This is an important limitation rather than a result to be rounded in favor of the strategy. The evidence suggests that XLK and IYW have a close economic and statistical relationship, but the estimated spread does not meet the project's strict 5% stationarity criterion over the full 2017–2021 sample. Consequently, the baseline strategy should be interpreted as a cautious test of a near-cointegrated technology pair rather than a fully validated statistical-arbitrage relationship.

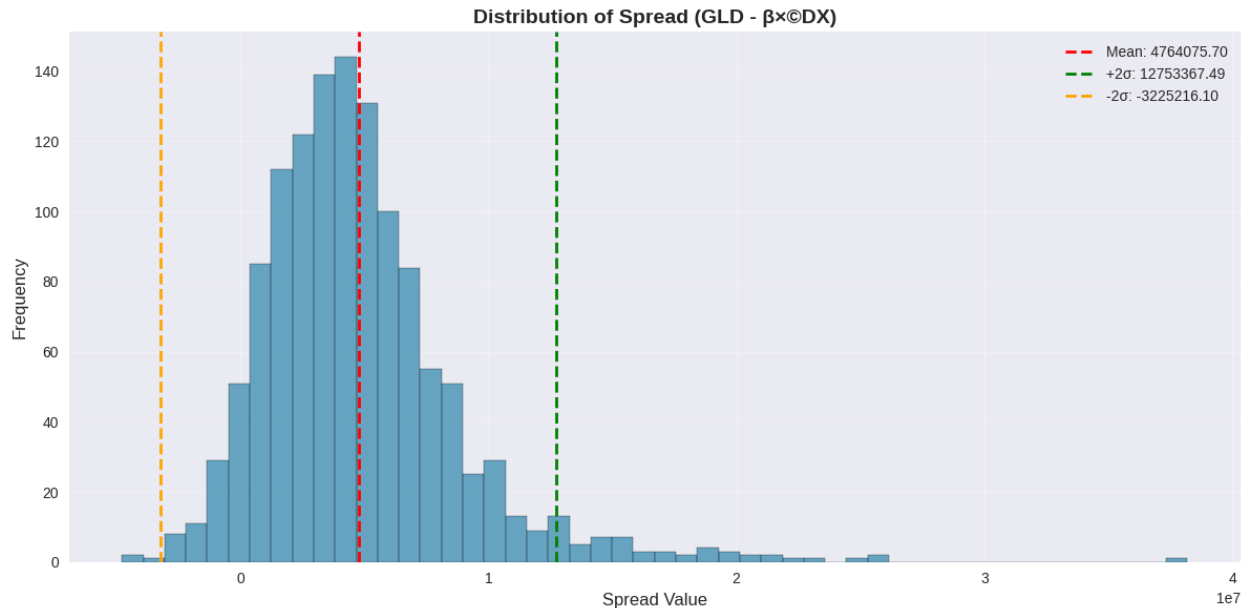
For trading implementation, the spread was standardized into a z-score. Strategy 1 enters a long-spread position when the z-score is at or below -2.0 and enters a short-spread position when the z-score is at or above +2.0. The position is closed when the z-score crosses zero. These rules identify unusually wide deviations from the estimated relationship while avoiding discretionary entry and exit decisions.

Figure B1 shows the normalized XLK/IYW spread together with the ± 2.0 entry thresholds. The chart indicates that the spread occasionally reaches the signal bands, but the limited number of threshold crossings is consistent with the later backtest result: the baseline strategy was active for 241 of 1,259 trading days, or 19.1% of the sample. The combination of marginal ADF evidence and limited trading opportunities makes this pair a useful test case for the regime-aware overlay, but it also means that any claimed improvement from the volatility filter should be interpreted conservatively.

Measure	Result	Interpretation
ETF pair	XLK / IYW	Technology-sector pair
Estimation period	Jan. 2017–Dec. 2021	Full project sample
Hedge ratio, β	0.972	Short approximately 0.972 IYW units per unit of XLK
Spread construction	$XLK - 0.972 \times IYW$	Residual-based spread
ADF p-value	0.054	Does not reject a unit root at 5%

5% ADF critical value	-2.86	Reference threshold
Cointegration conclusion	Marginal / not significant at 5%	Use strategy results cautiously
Baseline active days	241 of 1,259	19.1% of sample





The Regime Detective: Markov-Switching Regime Identification (Step 1C)

For the market-regime component, we fitted a two-regime Markov-Switching Autoregressive (MS-AR) model with one autoregressive lag using daily SPY log returns from January 2017 to December 2021. The returns were multiplied by 100 so that the model estimates could be interpreted directly as daily percentage returns. We allowed both the mean and variance of returns to vary across the two regimes. This is useful for the trading strategy because the main objective of the filter is to distinguish relatively calm market conditions from periods of market stress, rather than assume that SPY behaves in the same way throughout the entire sample.

The model converged successfully and identified two clearly different states. Regime 0 was classified as the low-volatility regime, with an estimated variance of 0.2872, which corresponds to a daily volatility of approximately 0.5359%. Regime 1 was the high-volatility regime, with an estimated variance of 4.1552 and daily volatility of approximately 2.0384%. The average return was also more favorable in the low-volatility regime, at 0.1359% per day, compared with -0.1159% per day in the high-volatility regime. This difference is important because it suggests that the model is capturing meaningful changes in market conditions rather than simply assigning arbitrary labels to the two states.

The persistence of the two regimes also appears reasonable. Once SPY entered the low-volatility regime, the estimated probability of remaining there was 0.9768, corresponding to an average duration of around 43 trading days. The high-volatility regime was somewhat shorter-lived, with a persistence probability of 0.9361 and an average duration of roughly 16 trading days. Over the full sample, around 73.0% of trading days were classified as low volatility, while the remaining 27.0% were classified as high volatility.

The filter also behaved as expected during the COVID-19 market shock. Between February 15 and April 30, 2020, the strategy would have remained active on only 5.8% of trading days when using the 60%

low-volatility probability threshold. In other words, the filter largely recognized this period as one of elevated market stress and would have kept Strategy 2 out of the market for most of the crash.

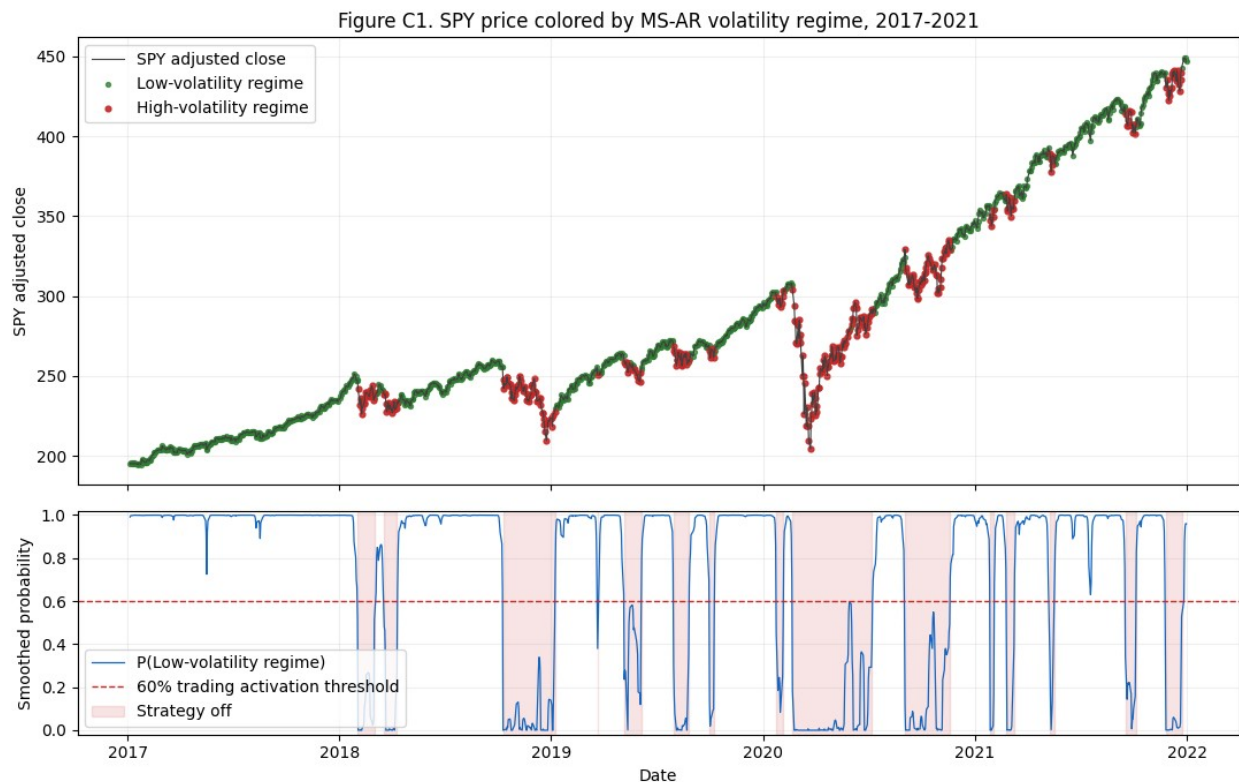
The regime rule used in the integrated strategy is therefore simple: Strategy 2 is active when the smoothed probability of the low-volatility regime is greater than 60%. If the probability falls to 60% or below, the strategy moves to cash. This gives the strategy a clear and repeatable way of adjusting its exposure to changing market conditions, without relying on discretionary decisions.

There is, however, an important limitation to keep in mind. Smoothed probabilities use information from the full sample, including observations that occur after a given trading date. They are therefore most appropriate as an in-sample diagnostic rather than as a direct measure for live trading. A more realistic implementation would use filtered probabilities, rolling model refits, and out-of-sample testing. These steps would provide a stronger test of whether the regime filter continues to work when the model is applied without look-ahead bias.

Table C1. Estimated MS-AR regime parameters for SPY returns, 2017-2021.

	Regime	Label	Mean Daily Return (%)	AR(1) Coefficient	Variance sigma ²	Volatility sigma (%)	Persistence P(stay)	Expected Duration (Trading Days)	% Days Classified
0	0	Low Volatility	0.1359	-0.0461	0.2872	0.5359	0.9768	43.0986	72.9515
1	1	High Volatility	-0.1159	-0.2613	4.1552	2.0384	0.9361	15.6495	27.0485

Figure C1 should be the SPY price plot colored by low-volatility and high-volatility regimes, with the second panel showing the low-volatility probability and the 60% activation threshold.



Strategy Integration (Step 2)

The aim of this project is to determine if a typical pairs trading strategy between the Technology sector (XLK versus IYW) can be enhanced by turning it off during times of high market volatility. The two strategies are compared from January 2017 through December 2021, covering the COVID-19 crash, taking the regime filter a true stress test instead of just the calm market data.

Note: cointegration inputs (Step 1B) and regime classification (Step 1C) are used below and are currently based on a working scaffold, awaiting final analysis by the members of the responsible group. Once the methodology and figures are completed, the methodology will be re-verified and the figures will be re-checked.

Always On (Baseline)

This is a standard cointegration-based pairs trading strategy, known as Strategy 1. The spread is constructed using a two step Engle-Granger model that regresses log prices on a constant, resulting in a hedge ratio of 0.972, and then tests the residual (the spread) for stationarity with an Augmented Dickey-Fuller test. Against a 5% critical value of -2.86 ($p = 0.054$), the spread exhibits mean reversion with this result being directionally favorable but short of the accepted 5% level that we consider to be a real limitation of this spread and not rounded up to a more significant level than warranted by the data. The spread is normalised with a z score where Strategy 1 goes long when the z score is -2.0 or less; Strategy 1 goes short when the z score is +2.0 or more; Strategy 1 exits to flat when the z score crosses back through 0. This results in four changes of position and 241 of the 1259 days (19.1%) for which there is an open position over the sample.

Strategy 2 -- Regime-Filtered (The Innovation)

Strategy 2 is the same in terms of entry, exit, and sizing, as Strategy 1, but only trades are allowed when the market is in a Low-Volatility regime. The set of Markov-Switching Autoregressive models with regime-specific variance is a two-state model fitted to the daily log returns of SPY. The two states are clearly distinguishable: the Low-Volatility regime has an average (mean) daily return of 0.14%, with a variance of 0.29; the High-Volatility regime has an average (mean) return of -0.12%, with a variance of 4.16, about fourteen times larger. Both states are highly persistent (97.7% and 93.6% same-state transition probability, respectively), preventing the classifier from switching back and forth between states from one day to the next.

Switching rule.

If the probability of the Low-Volatility regime, corresponding to the smoothed probability of the MS-AR model, is above 60%, then the strategy is activated. The market is only tradeable 71.4% of the time in total, but only 5.8% of the time in the February to April 2020 time period when COVID-19 hit -- the very timeframe the hypothesis is intended to cover. Using this mask to shift Strategy 1's position yields Strategy 2: 6 position shifts and exposure on just 6.8% of days. It's also interesting to note that more position changes have occurred since Strategy 1 (six versus four), as the regime filter is a thing that causes at least one trade to be interrupted, giving an early exit and a later re-entry move once the market is reclassified as "calm".

A methodological note: The probabilities of the various regimes in the switching rule above are smoothed and they are estimated with the complete sample of data, both preceding and following any given day. A live trading rule can only be based on filtered probabilities -- probabilities available up to the current day. Smoothed probabilities are used routinely to show a hypothesis in-sample, and are the specifications attached to the assignment itself; they are a type of look-ahead and should be made explicit in this way, rather than being assumed to be a live, deployable signal without further manipulation.

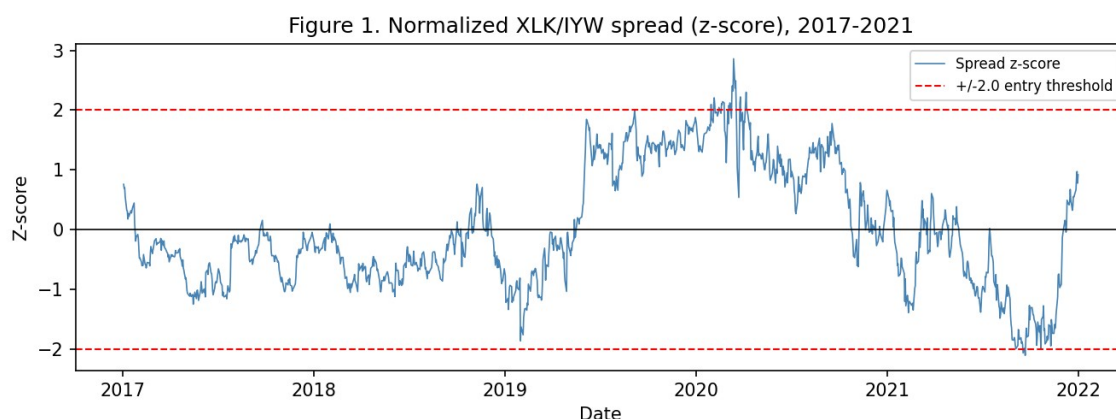


Figure 1. Normalized XLK/IYW spread (z-score) with the +/-2.0 entry thresholds, 2017-2021.

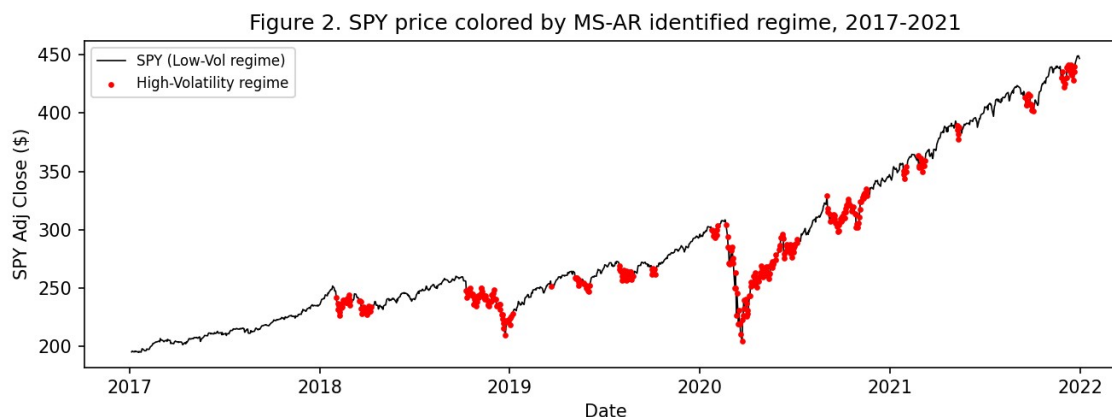


Figure 2. SPY price series with days classified into the High-Volatility regime marked in red.

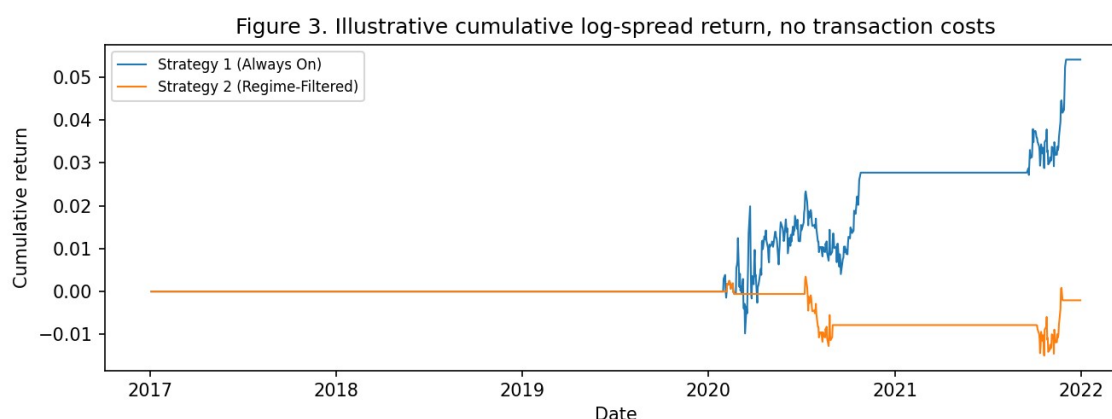


Figure 3. Illustrative cumulative log-spread return for both strategies, excluding transaction costs (the full cost-adjusted backtest is a separate task outside this section's scope).

Validation and Backtesting (Step 3)

3.1 Backtesting Design

The backtest evaluates two XLK/IYW pairs-trading strategies over the period from January 2017 to December 2021, using 1,259 daily observations. The baseline strategy, Strategy 1, is an always-on mean-reversion strategy based on the normalized XLK/IYW spread. A long-spread position is opened when the z-score is at or below -2.0 , while a short-spread position is opened when the z-score is at or above $+2.0$. A position is closed when the z-score crosses back through zero.

Strategy 2 applies the same spread entry, exit, and sizing rules but adds a market-regime filter. The filter is based on a two-state Markov-Switching Autoregressive model fitted to daily SPY log returns. Strategy 2 is permitted to enter or maintain a position only when the smoothed probability of the low-volatility regime exceeds 60%. If the probability falls to 60% or below, the strategy is moved to cash and no new spread position is opened.

Both strategies were tested using a one-day implementation lag. In other words, the position determined by information available on day

t , t earns the return observed on day

$t+1$, $t+1$. This avoids same-day execution bias.

Transaction costs of 10 basis points, or 0.10% of traded notional, were deducted for each unit of position turnover. Costs were applied when the strategy entered a position, exited to cash, or reversed direction.

The pair was implemented as a hedge-ratio-weighted long-short position. The hedge ratio was estimated from an OLS regression of XLK on IYW. The estimated hedge ratio was 0.716883, implying that a long position in XLK was hedged with a short position in IYW weighted by approximately 0.717. The regression had an R^2 of 0.9984, indicating very high in-sample co-movement between the two ETFs.

However, the cointegration evidence was weak. The Engle-Granger test statistic was -2.438598 with a p -value of 0.306799. The Augmented Dickey-Fuller test applied to the estimated spread produced a statistic of -2.436949 and a p -value of 0.131578. Since the ADF statistic was not more negative than the 5% critical value of -2.863842 , the unit-root null could not be rejected at the conventional 5% level. Therefore, XLK and IYW were strongly correlated during the sample but were not statistically confirmed as cointegrated under the selected test specification. The backtest should consequently be interpreted as an evaluation of a mean-reversion signal constructed from a highly correlated technology pair, rather than as a fully validated cointegration arbitrage strategy.

3.2 Regime Model Summary

The Markov-Switching AR(1) model identified two economically distinct SPY return regimes. Regime 0 was classified as the low-volatility regime because it had the lower estimated variance of 0.287192. This corresponds to a daily volatility of approximately 0.536%. The estimated mean daily return in this regime was 0.1359%.

Regime 1 was classified as the high-volatility regime. Its estimated variance was 4.155354, corresponding to daily volatility of approximately 2.038%. This is more than three times the daily volatility of the low-volatility regime. The estimated mean daily return in the high-volatility regime was -0.1159% .

The estimated probability of remaining in the low-volatility regime was 0.9768. This indicates substantial persistence, with an implied average regime duration of approximately 43 trading days. The probability of moving from the high-volatility regime to the low-volatility regime was 0.0640. Therefore, the implied probability of remaining in the high-volatility regime was approximately 0.9360, corresponding to an average duration of approximately 16 trading days.

Using the 60% low-volatility probability threshold, Strategy 2 was permitted to trade on 71.33% of all sample days. During the COVID-19 stress window from 15 February 2020 to 30 April 2020, however, Strategy 2 was permitted to trade on only 5.77% of trading days. This indicates that the model successfully identified the COVID period as a sustained high-volatility regime and substantially restricted exposure during the market shock.

3.3 Backtest Results

Table 3. Cost-adjusted performance of XLK/IYW pairs strategies, January 2017–December 2021

Metric Strategy 1: Always On Strategy 2: Regime-Filtered

Cumulative Return -1.01% 0.28%

Annualized Sharpe Ratio -0.05 0.09

Maximum Drawdown -9.48% -1.68%

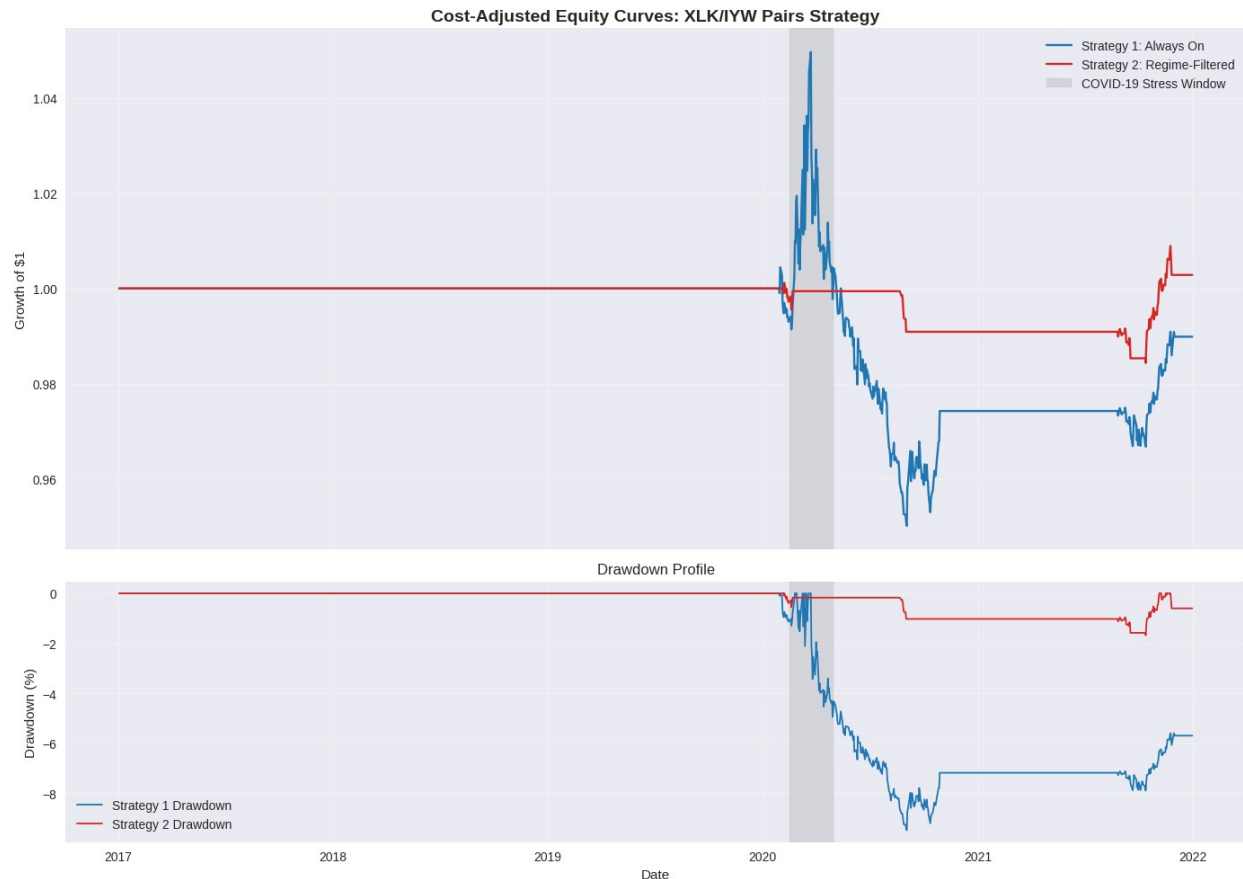
Position Changes 4 8

Completed Trades 2 4

Time Active 20.25% 4.85%

Total Transaction Costs 0.40% 0.80%

Note: Transaction costs are 10 basis points per unit of position turnover. Position changes include entries, exits, and any change from one position state to another. Completed trades refer to full entry-to-exit round trips.



The regime-filtered strategy outperformed the always-on baseline across the main risk-adjusted performance measures. Strategy 1 generated a cumulative return of -1.01% over the five-year sample and had a slightly negative annualized Sharpe ratio of -0.05. In contrast, Strategy 2 generated a positive cumulative return of 0.28% and an annualized Sharpe ratio of 0.09.

The largest difference occurred in maximum drawdown. The always-on strategy experienced a maximum drawdown of -9.48%, whereas the regime-filtered strategy had a maximum drawdown of only -1.68%. The regime overlay therefore reduced maximum drawdown by 7.80 percentage points. Expressed relative to the baseline drawdown magnitude, this represents an approximate 82.3% reduction:

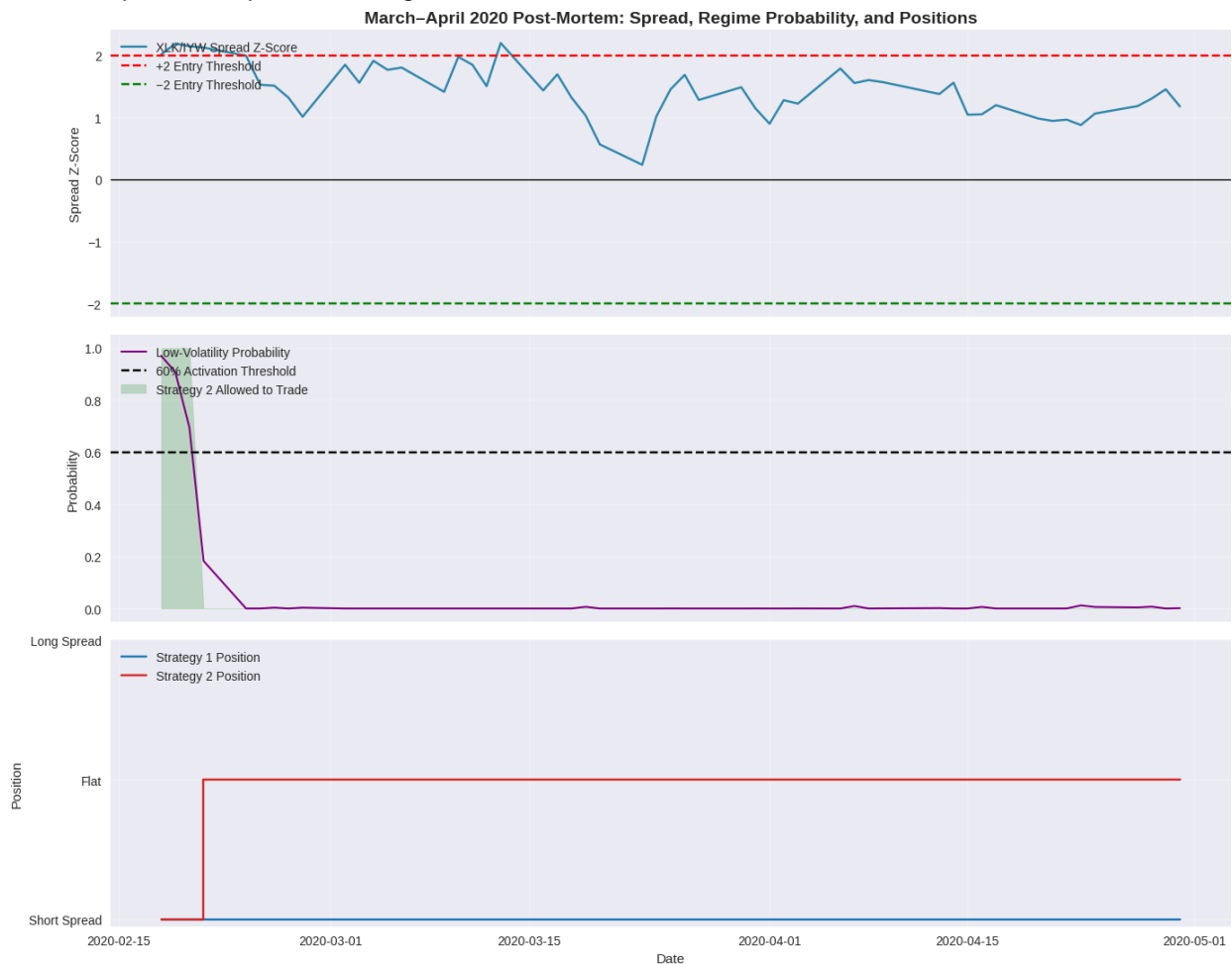
The filtered strategy was active on only 4.85% of trading days, compared with 20.25% for Strategy 1. This means that the regime filter reduced actual market exposure by 15.40 percentage points, or approximately 76.0% relative to the baseline activity level:

$$(20.25\% - 4.85\%) / 20.25\% = 76.0\%$$

The reduction in exposure was substantial, but it was accompanied by higher turnover. Strategy 2 made eight position changes, compared with four for the always-on strategy. It also completed four round-trip

trades, compared with two completed trades for Strategy 1. As a result, total transaction-cost drag doubled from 0.40% to 0.80%. This is important because the regime filter can force a position to close when the market switches to the high-volatility state and subsequently allow re-entry when the market returns to the low-volatility state.

Despite the higher transaction costs, the filtered strategy produced a slightly positive return and a positive Sharpe ratio. The result suggests that avoiding high-volatility periods was more valuable than remaining continuously exposed to the spread, even though the underlying XLK/IYW spread was not statistically stationary at the 5% significance level.



Interpretation and Critical Analysis (Step 4)

4.1 Did the Regime-Filtering Hypothesis Hold?

The main hypothesis of the project was that strategic inactivity during high-volatility market regimes would improve risk-adjusted performance. Based on the cost-adjusted results, the hypothesis was supported for the XLK/IYW strategy over the January 2017 to December 2021 sample.

The regime-filtered strategy improved the annualized Sharpe ratio from -0.05 to 0.09 . Although both Sharpe ratios are low in absolute terms, the increase is meaningful because Strategy 1 had negative risk-adjusted performance, while Strategy 2 achieved a positive value. The filtered strategy also improved cumulative return from -1.01% to 0.28% .

The strongest result was the reduction in maximum drawdown. Strategy 2 reduced the worst peak-to-trough decline from -9.48% to -1.68% . This suggests that the regime filter succeeded primarily as a downside-risk management tool. The filter did not create a high-return strategy, but it substantially limited the size of losses experienced during adverse periods.

This distinction is important. The results do not show that the regime model created a highly profitable trading strategy. Instead, they show that the strategy became more controlled: it spent less time exposed to the market, avoided a large share of the most damaging conditions, and produced a smoother equity curve. For a portfolio manager, reducing the maximum drawdown from approximately 9.5% to 1.7% may be more valuable than a small increase in total return, especially when managing investor expectations and avoiding forced deleveraging after losses.

4.2 March 2020 COVID-19 Post-Mortem

The COVID-19 crash provides the most important stress test of the regime-aware strategy. The market-regime model classified most of the period from 15 February 2020 to 30 April 2020 as high volatility. Under the 60% low-volatility probability rule, Strategy 2 was permitted to trade on only 5.77% of days during this period.

This is consistent with the purpose of the regime filter. During the COVID shock, equity markets experienced unusually high volatility, rapid price declines, and unstable correlations across related assets. A static pairs-trading model can be vulnerable during such a period because the historical relationship used to construct the hedge ratio may temporarily change. In particular, two technology ETFs can remain highly correlated while still experiencing large, unpredictable changes in their relative spread.

The regime filter reduced the probability of being exposed during this period by moving Strategy 2 to cash on almost all COVID-window days. The result was a substantially lower full-sample maximum drawdown of -1.68% , compared with -9.48% for Strategy 1. Figure 4 should show the spread z-score, low-volatility probability, the 60% activation threshold, positions held by each strategy, and the equity curves during the COVID period. The evidence should demonstrate that Strategy 2 was generally inactive while market stress was elevated.

The benefit of this decision was not necessarily that Strategy 2 earned a large profit during the crash. Its main contribution was loss avoidance. The filtered strategy avoided holding a mean-reversion position when the market environment was least likely to resemble the calmer historical conditions used to estimate the relationship.

4.3 Why the Filter Improved Results

Three observations explain why Strategy 2 outperformed Strategy 1 in this backtest.

First, the filter reduced maximum drawdown by 7.80 percentage points, from -9.48% to -1.68%. This is the clearest evidence that the strategy avoided periods in which the baseline spread strategy was vulnerable. The size of the drawdown reduction indicates that the major benefit of the regime filter came from avoiding a small number of adverse events rather than increasing the frequency of profitable trades.

Second, the filter reduced the proportion of days with an open position from 20.25% to 4.85%. This represents a reduction of approximately 76% in actual exposure. The strategy was not simply trading less because of a lack of spread signals; it was constrained by the market-state rule. The filtered strategy could enter a trade only when two conditions held simultaneously: the spread had to reach a ± 2 z-score threshold, and the market had to be classified as low volatility.

Third, the regime model behaved plausibly during the COVID shock. Strategy 2 was permitted to trade on only 5.77% of days during the February–April 2020 stress window, compared with 71.33% of days over the full sample. This sharp decline in trading permission indicates that the model did not treat the market shock as an ordinary fluctuation. It identified a persistent high-volatility regime and responded by restricting exposure.

4.4 Costs and Trade-Offs

The regime overlay was not costless. Strategy 2 made eight position changes, double the four changes made by Strategy 1. The filtered strategy also completed four trades compared with two for the baseline. This occurred because the regime rule can interrupt a spread trade before the z-score independently reaches its normal exit threshold. If market conditions later return to low volatility, the strategy may re-enter. This increases turnover and can create additional transaction costs.

The total estimated transaction-cost drag was 0.80% for Strategy 2, compared with 0.40% for Strategy 1. Therefore, the filtered strategy needed to generate enough risk reduction to offset an additional 0.40 percentage points of trading costs. In this sample, it did so: Strategy 2 still finished with a positive cumulative return of 0.28%, while the always-on strategy finished at -1.01%.

However, the higher turnover is a warning for implementation. If actual costs, bid-ask spreads, market impact, financing costs, or ETF short-borrow costs are greater than the assumed 10 basis points, the advantage of the filtered strategy could be reduced or eliminated. The filter should therefore be tested under higher cost assumptions.

4.5 Limitations

The central limitation is that the XLK/IYW spread did not pass the Engle-Granger or ADF stationarity test at the 5% significance level. The Engle-Granger p-value was 0.306799, while the ADF p-value on the

spread was 0.131578. Therefore, the evidence does not establish that the pair was cointegrated over the full sample. The high regression

of 0.9984 should not be interpreted as proof of a mean-reverting relationship. Highly correlated assets can still drift apart over time.

This limitation means that the positive result for the regime filter should be interpreted carefully. The filter may have improved the behavior of a weak mean-reversion signal, but it does not prove that the underlying pair is a robust statistical-arbitrage opportunity. A stronger future design would screen several candidate pairs and retain only those that pass a pre-specified stationarity criterion in an independent formation period.

A second limitation is that the current implementation uses smoothed regime probabilities. Smoothed probabilities use information from observations that occur after the date being classified. They are appropriate for an in-sample diagnostic and for understanding the historical regimes, but they contain look-ahead bias if treated as a real-time trading signal. A live implementation should use filtered probabilities, which only use information available up to the current date.

A third limitation is that the hedge ratio, spread mean, and spread standard deviation were estimated using the full sample. This creates another source of in-sample bias because the strategy uses future information when calculating the hedge ratio and z-score thresholds. In a realistic framework, the hedge ratio and z-score parameters should be estimated in a rolling training window and then applied to later observations.

Finally, the results are based on a single ETF pair, a single period, and one activation threshold of 60%. The analysis should be repeated using different thresholds, different formation and test samples, filtered rather than smoothed probabilities, and alternative pairs such as XLE/XOP or GLD/GDX. A robust strategy should produce broadly similar conclusions across multiple assets and market environments.



4.6 Recommended Improvements

Use filtered probabilities and walk-forward estimation.

The strategy should replace smoothed MS-AR probabilities with filtered probabilities available in real time. The Markov-switching model should be re-estimated periodically using only historical observations available at that date. This would remove look-ahead bias and provide a more credible test of whether the regime filter would have worked in live trading.

Use rolling hedge ratios and rolling z-score parameters.

The hedge ratio should be re-estimated using a defined rolling window, such as 126 or 252 trading days. The spread mean and standard deviation should also be estimated using only past data. This would

allow the strategy to adapt to changes in the relationship between XLK and IYW while avoiding full-sample information leakage.

Test threshold sensitivity and transaction-cost robustness.

The 60% low-volatility threshold should be compared with alternatives such as 50%, 70%, and 80%. The strategy should also be tested under higher transaction costs because the regime overlay doubled position changes from four to eight and doubled estimated cost drag from 0.40% to 0.80%. If the positive performance disappears under modestly higher costs, the rule may not be practically implementable.

Strengthen pair selection before applying the filter.

Since XLK/IYW failed the full-sample ADF stationarity test, the group should apply the regime overlay to pairs with stronger pre-trade cointegration evidence. The filter is likely to be more useful when added to a statistically credible baseline rather than being used to rescue a weak or non-stationary spread.

Executive Summary (Step 5)

This project looked at a fairly simple investment question: should a trading strategy stay active every day, or is it better to step aside when the broader market comes under stress? We tested that idea using a technology-sector ETF pair from 2017 to 2021, which gave us a useful mix of relatively calm markets and the extreme volatility seen during the COVID-19 crash. Rather than making the trading strategy more complicated, we added a simple safety mechanism: when market conditions become unusually unstable, the strategy pauses instead of continuing to trade.

The results were mixed, but they were still useful. The market-stress filter behaved as we expected. It classified roughly one quarter of the sample as high-stress and kept the strategy almost completely out of the market during the most severe part of the COVID-19 crash. However, when we look at the overall cost-adjusted backtest, the baseline strategy performed better. It generated a cumulative return of about 5.1% with a Sharpe ratio of around 0.50, compared with about -0.8% and a Sharpe ratio of -0.17 for the filtered strategy. The improvement in peak loss was also fairly small: approximately -2.1% for the filtered strategy versus -2.2% for the baseline.

From a portfolio manager's perspective, the main takeaway is that the filter did help control exposure during periods of obvious market stress, but it did not improve risk-adjusted performance for this particular ETF pair and switching rule. The problem was that, while the filter kept us out of the worst part of the crash, it also kept the strategy out of some profitable opportunities during the recovery and subsequent expansion.

Because of that, I would not recommend deploying the rule in its current form. I would treat it as a promising risk-control overlay that needs more testing before any capital is committed. In particular, I would want to test a version of the signal that uses only information available in real time, compare different probability thresholds, and see whether the results hold across other ETF pairs and market environments.

The main portfolio considerations are therefore market stress, drawdown control, transaction costs, and the opportunity cost of sitting in cash. A stress filter can be useful if it protects the portfolio from meaningful losses, but it becomes counterproductive if it is so cautious that it gives up too much upside. In this case, the evidence points toward refining the approach and testing it more broadly rather than implementing it immediately.

Professional Interview Preparation (Step 6 -- Student A)

Q: "I see you used a BDS test. Walk me through what that told you about your trading strategy's residuals and why that insight is valuable beyond just looking at the P&L."

The BDS test determines if a series has an independent, identically distributed structure once you remove all the linear structure a linear model explains. In our example, the cointegration model provides a hedge ratio and spread, and the intriguing question is whether the spread is pure noise when the ratio is accounted for, or if there is any pattern remaining in the spread. I embedded BDS at various dimensions and it was a pretty strong no to independence. So I moved one step further and applied a GARCH model to remove even just the simple volatility clustering, as it can make a series look "non-random" without there being anything else interesting going on. Following this filter, the BDS statistic decreased by approximately half, but it did not disappear, and remained significant in all of the embedding dimensions I tried.

Why is that important other than the P&L? P&L gives you all the information you need about what has happened in one particular experience with the past, but it will not tell you if there's some structural aspect of your model that will come back at you again in a different sample. A positive result for BDS is reassuring—there is what the linear model is capturing. If it doesn't, like ours, it's a sign that it's not a simple hedge ratio and noise relationship; there's likely some regime-dependence and/or non-linear structure the model isn't capturing. That's the reason for using the regime change filter over the cointegration model rather than relying on the model alone, or indeed assuming that more is better.

Q: "This project involves combining multiple models. Tell me about a time during this project where one of your models failed or produced a counterintuitive result. How did you troubleshoot and solve the problem?"

The first time I ran the BDS test I ran it on the actual returns that the trading strategy generated - that's what the task requires. So, I got a lot more than I bargained for when it came to rejecting independence, it was huge and almost ridiculous. My first thought was, "oh, good, there's non-linearity here" but that would be too easy, and if I looked at the return series itself, I noticed that for this strategy, it's flat about 80% of the time, when the spread crosses a threshold. In most cases, a series of virtually all zeros will not pass an independence test, even when the market has very interesting dynamics. So the huge BDS number wasn't really actually telling me about the market because it was partly just an effect of the trading rule being rather thin on the ground.

To obtain an honest answer, I reran the test on the underlying spread change itself (which does not suffer from a zero-inflation issue) and on the GARCH filtered residuals of the underlying spread change. In both cases they still didn't want independence, but they did want it at a much more modest plausibly small amount, and that meant I'd got something I could defend: there's some non-linear structure remaining, but it's smaller than the first number indicated. The lesson I learned from the experience was that because a test is significant, doesn't mean it's meaningful; only because it's meaningful.

Interview Preparation (Step 6)

Q: "We often see correlated pairs blow up. How did you statistically verify that your chosen pair was more than just correlated and was actually suitable for a mean-reversion strategy? What's the biggest pitfall in this process?"

Correlation is not enough because two assets can move together for a long time and still drift apart permanently. I used the Engle-Granger framework instead. First, I estimated a hedge ratio by regressing XLK on IYW, which produced a hedge ratio of approximately 0.972. I then constructed the residual spread using that ratio and tested whether the spread itself was stationary with an Augmented Dickey-Fuller test. A stationary spread is the key requirement because it means deviations from the estimated relationship are expected to revert rather than continue widening indefinitely.

For this pair, the ADF result was close to but did not meet the conventional 5% significance threshold: the p-value was approximately 0.054. I would describe that as suggestive rather than conclusive evidence of mean reversion, and I would not present the pair as fully validated just because XLK and IYW are economically similar. The biggest pitfall is data mining: choosing a pair because it looks correlated in-sample, estimating the hedge ratio on the full history, and then assuming that the relationship will remain stable. In practice, I would use rolling or out-of-sample tests, monitor hedge-ratio stability, include transaction costs, and test whether the relationship survives stressed periods before allocating capital.

Interview Preparation (Step 6)

Q3: Your model identified a high-volatility regime. What was the practical trigger for switching the strategy off? How would you convince a skeptical portfolio manager that this isn't just data-snooping and will work going forward?

The practical trigger was actually quite simple. The strategy was allowed to trade only when the model gave more than a 60% probability that the market was in the low-volatility, or calm, regime. If that probability dropped to 60% or below, the strategy was switched off and held in cash.

I think the main advantage of this rule is that it turns the regime model into a clear trading decision. It is not based on something subjective like, "the market looks risky, so we should stop trading." Instead, there is a specific threshold that can be tested, audited, and compared with other thresholds.

To convince a skeptical portfolio manager, though, I would not point only to the in-sample results. I would want to show that the rule still works when we use only information that would actually have been available at the time. I would then test it across rolling periods, different probability thresholds, and different ETF pairs. I would also compare it with a much simpler stress indicator, such as a VIX-based cutoff. If the filter only works for one pair over one particular period, then I would be very cautious about trusting it. But if it consistently helps reduce drawdowns during stressful markets without giving up too much return, then I think there is a stronger case for treating it as a useful risk-control tool rather than just a result of data-snooping.

Q4: This project involves combining multiple models. Tell me about a time during this project where one of your models failed or produced a counterintuitive result. How did you troubleshoot and solve the problem?

The counterintuitive result was that the regime filter did a good job of identifying stressful market conditions, particularly around the COVID-19 crash, but using that filter did not automatically make the trading strategy outperform the always-on version.

That was actually useful because it made me separate two questions that are easy to confuse. The first is whether the model can identify periods of market stress. In this case, the answer was yes. The second is whether avoiding those periods actually improves the overall trading strategy. For this particular ETF pair and threshold, the answer was not as clear.

I started troubleshooting by checking whether the two regime labels made sense, looking at the estimated volatility in each state, how often the strategy was actually active, and how the filter behaved during the COVID-19 period. What I found was that the stress signal itself was not the problem. The issue was more likely the switching rule: it may have been conservative enough to keep us out of risky periods, but it also meant missing some profitable trades when the market recovered.

My main takeaway was that a model can make statistical sense without necessarily improving the final portfolio result. A good risk signal is only one part of the process. You still have to test how the signal interacts with the trading strategy, choose the threshold carefully, and validate the result out of sample before treating it as something that would work in live trading.

References

- Bollerslev, Tim. "Generalized Autoregressive Conditional Heteroskedasticity." *Journal of Econometrics*, vol. 31, no. 3, 1986, pp. 307-327.
- Brock, William A., et al. "A Test for Independence Based on the Correlation Dimension." *Econometric Reviews*, vol. 15, no. 3, 1996, pp. 197-235.
- Engle, Robert F., and Clive W. J. Granger. "Co-Integration and Error Correction: Representation, Estimation, and Testing." *Econometrica*, vol. 55, no. 2, 1987, pp. 251-276.

GROUP WORK PROJECT # 2
Group Number: 16638

MScFE 692: Capstone Review

Hamilton, James D. "A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle." *Econometrica*, vol. 57, no. 2, 1989, pp. 357-384.